

4. Use subtraction rule when needed.
- (a) 4 is written as IV (1 less than 5)
 - (b) 9 is written as IX (1 less than 10)
 - (c) 40 is written as XL (10 less than 50)
 - (d) 90 is written as XC (10 less than 100)
 - (e) 400 is written as CD (100 less than 500)
 - (f) 900 is written as CM (100 less than 1000)

Example :

$$\begin{aligned} (1) \quad 49 &= \underbrace{10 + 10 + 10 + 10}_{XL} + \underbrace{(10 - 1)}_{IX} \\ &= \quad \quad \quad XL \quad \quad IX \\ 49 &= XLIX \end{aligned}$$

[**Note** : In the Roman number system, any symbol is repeated a maximum of 3 times. However, people using this system were not always consistent.]



Exercises



NCERT Folder

Intext Questions

1. Imagine that we are living in the Stone Age, say, around ten thousand years ago. Suppose we have a herd of cows. Here are some natural questions that we might ask about our herd :

- (a) How do we ensure that all cows have returned safely after grazing?
- (b) Do we have fewer cows than our neighbour?
- (c) If there are fewer, how many more cows would we need so that we have the same number of cows as our neighbour?

We need to tackle these questions without the use of the number names or written numbers of the Hindu number system. How do we do it?

(Textbook page 51) (CBQ)

Ans.

- (a) We will keep one stick or pebble for each cow when they leave for grazing. When the cow came back, we will match the number of cows with the number of sticks or pebbles.

If the numbers are same, it means all cows have returned safely.

- (b) We can match our sticks / pebbles with our neighbour's sticks / pebbles. If our sticks finish first while our neighbour still has some left, then we have fewer cows.
- (c) We can compare the sticks / pebbles. The number of extra cows we need is the number of sticks / pebbles our neighbour has left after we finish matching ours.

[**Note** : A sample answer is given, but learners can come up with their own examples and write their own answers, as there are many possible solutions.]

2. How many numbers can you represent in this way using the sounds of the letters of your language? (Textbook page 53)

Ans. Do it yourself.

3. Do you see a way of extending this method to represent bigger numbers as well? How?

(Textbook page 53)

Ans. By looking at the table, we understand that